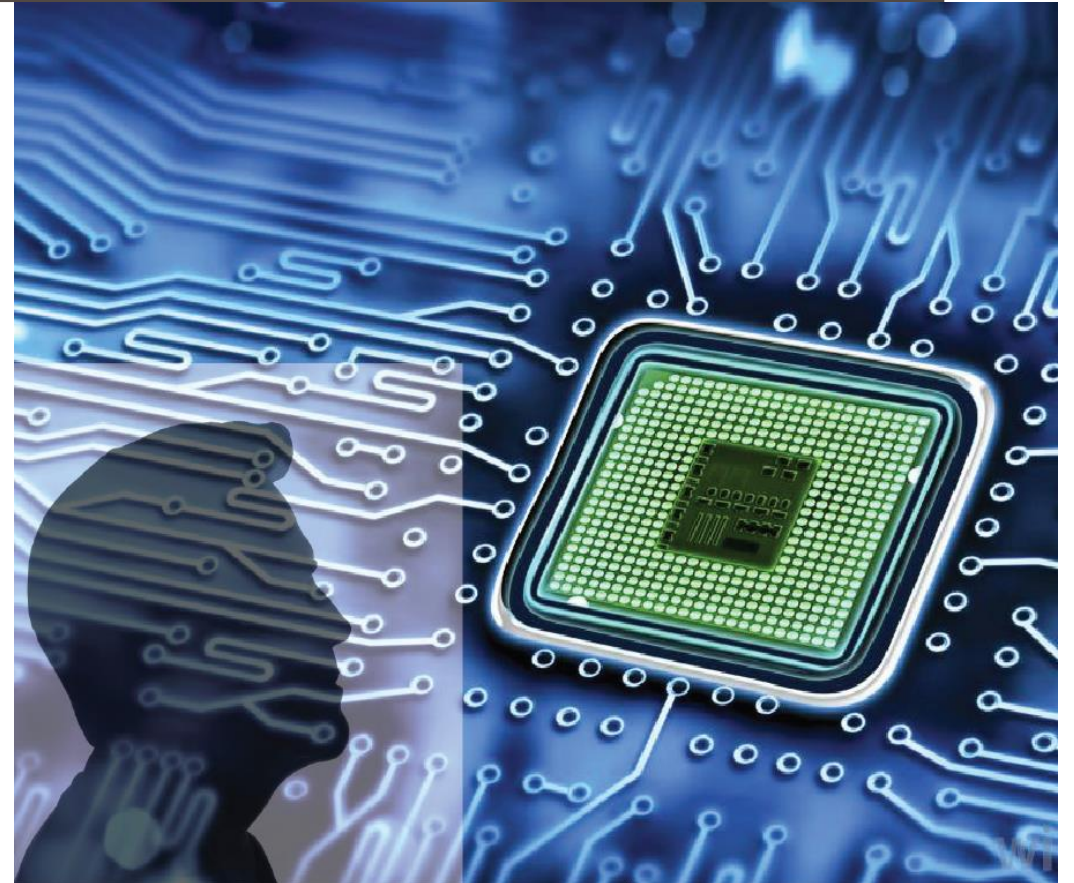


VLSI DESIGN FLOW

Week-13 Lecture-1 Physical Design

Sneh Saurabh
18th November, 2024

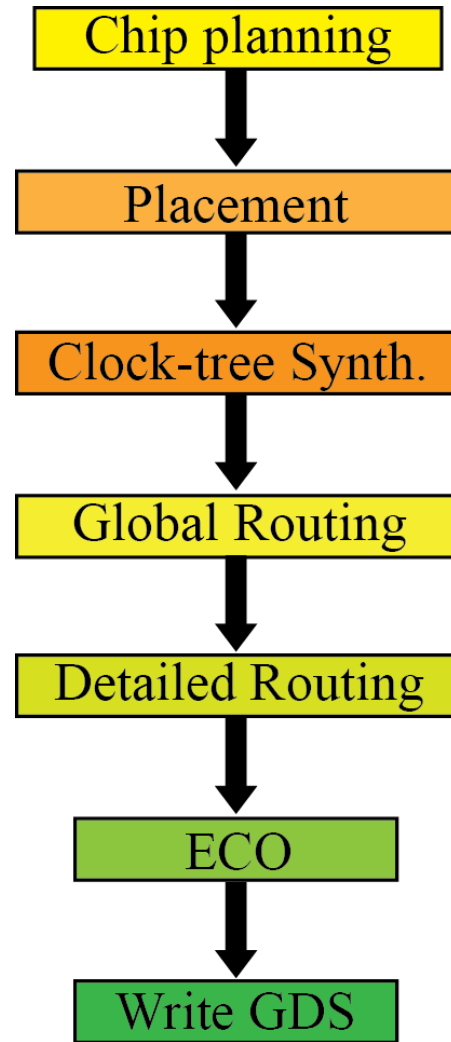


VDF: Revision/Announcements

-

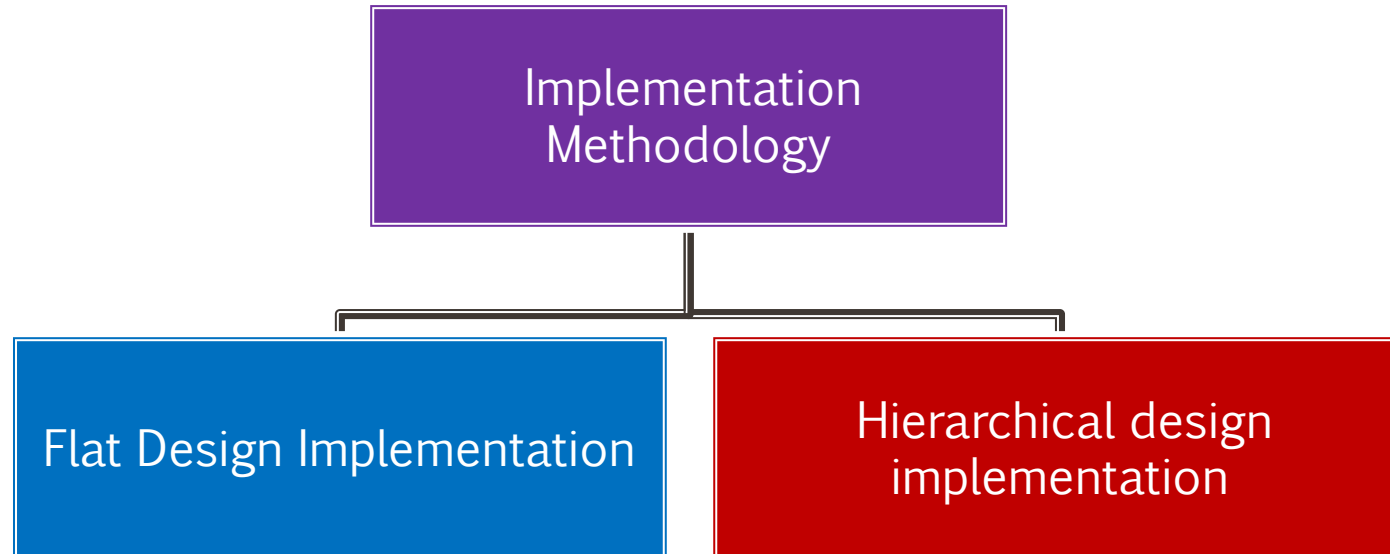


Physical Design



Chip Planning

Implementation Methodology



Tasks in hierarchical design implementation

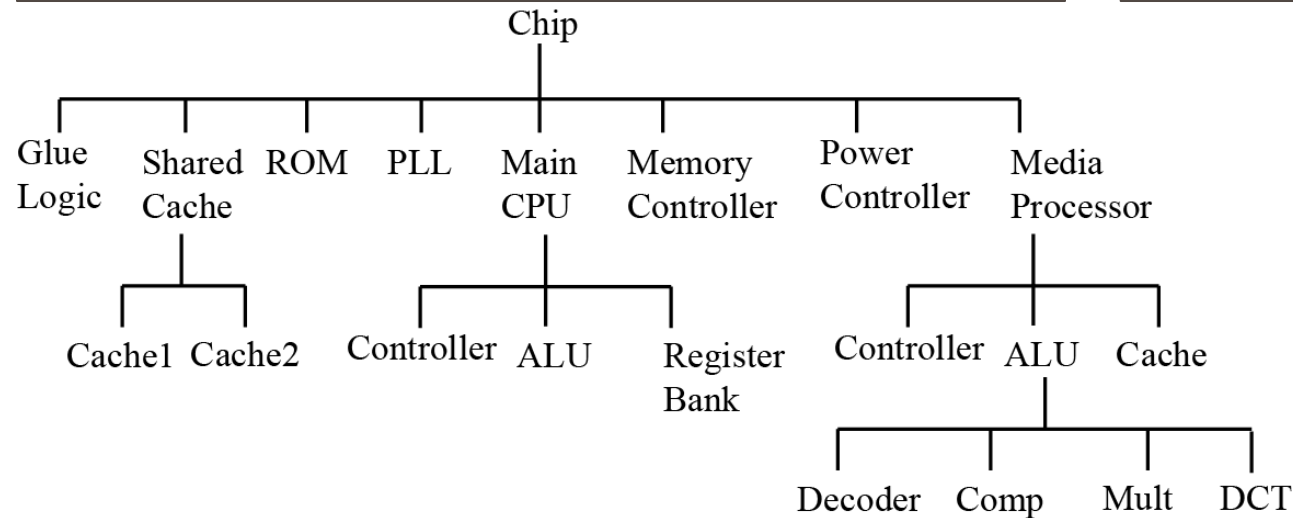
1. Partitioning
2. Budgeting
3. Block Implementation
4. Top-level Assembly

Partitioning

How to partition?

- Partition using logical functionality

- Partitions are called *blocks*

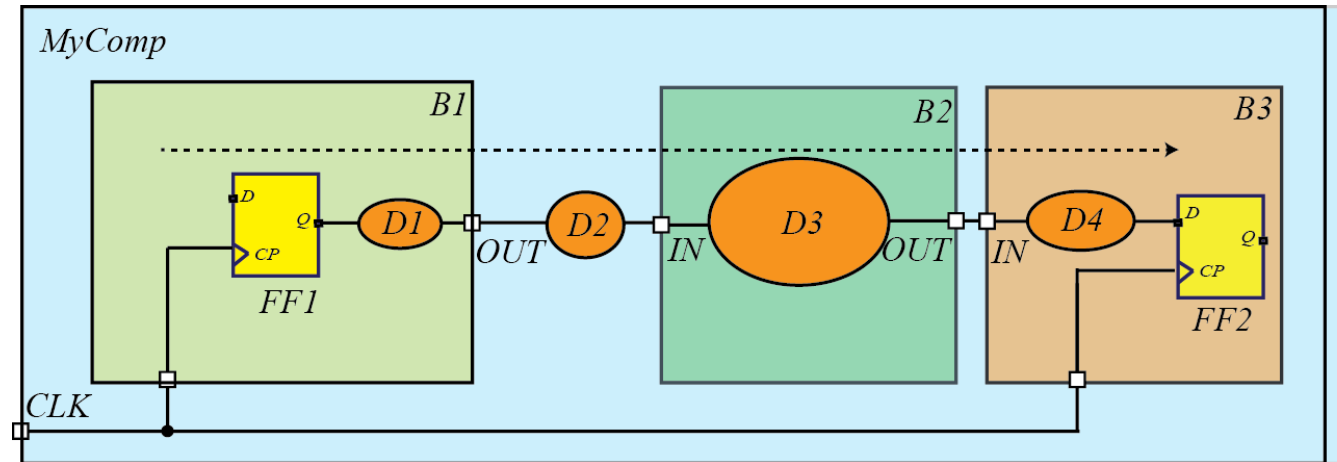


Other approaches:

- Group modules into clusters
- Partition a netlist using partitioning algorithm
 - Reduce the number of cuts or nets crossing blocks

Budgeting

- Process of allocating some fraction of a clock cycle to different blocks and the top-level design for signals crossing block boundaries



- Create block level SDC

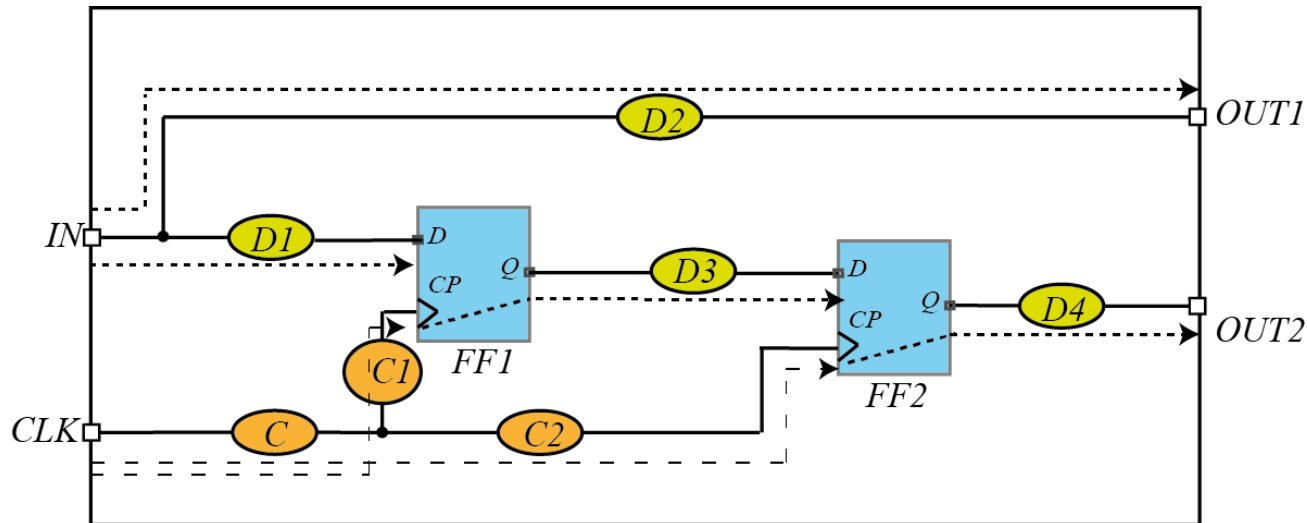
Block Implementation and Top-level Assembly

Block Implementation

- Implement each block separately
- Use allocated timing budgets and block constraints (verify at block level)

Top-level Assembly

- Integrate all the blocks at the top-level
- Carry out verification at the top-level
- Omit details of the timing paths contained entirely within a block



Abstract Timing Model of Blocks:

- Retain information of interface timing paths of the blocks
- Extracted Timing Model (ETM)
- Interface Library Model (ILM)

Hierarchical Design Methodology: Merits and Challenges

Advantages:

- Physical implementation and other EDA tools need to handle smaller problems
- Multiple blocks can be concurrently implemented by different teams
 - Overall design time decreases

Disadvantages:

- Challenging to partition a design optimally
- Lose some opportunities of inter-block optimizations

Floorplanning

Floorplanning: Basics

- Planning phase of the layout
 - Designer's intent about physical design
- Prepares a design for other physical design tasks
 - Huge impact on the final FoM
- Must consider routability, performance, power, etc.

Major Tasks:

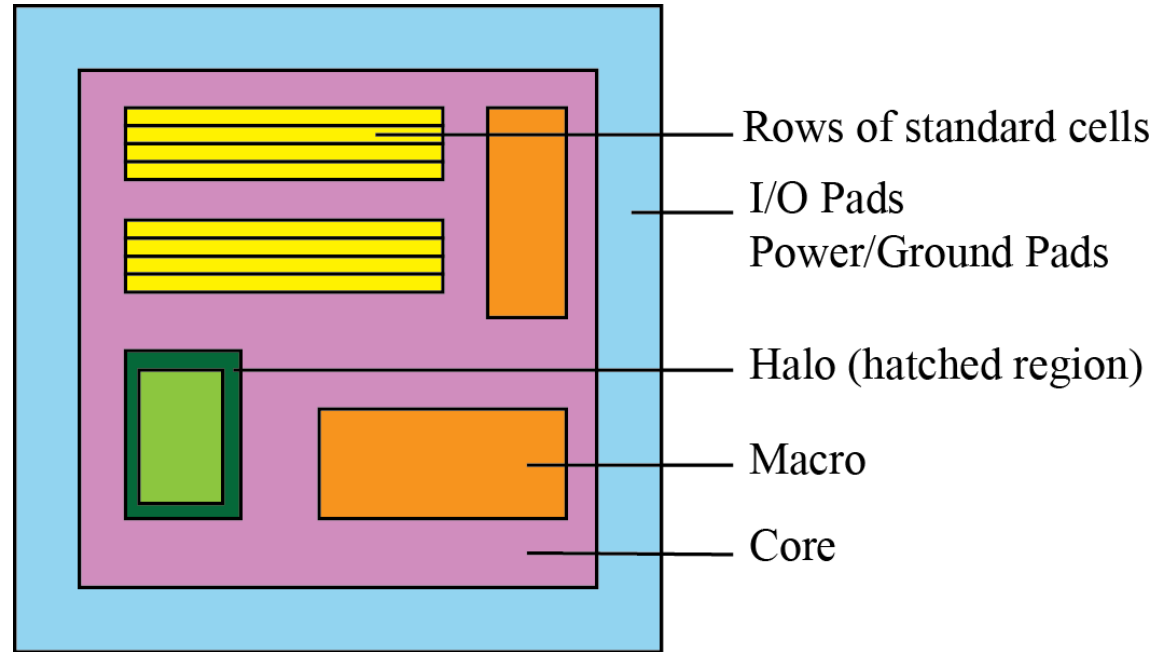
1. Define die/chip size and aspect ratio
2. I/O cell placement
3. Hard macros/block placement
4. Pin assignment
5. Create rows for standard cells

Die Size

Goal: choose smallest size of die that can fit the design

Die area should include area for:

- IO Cells/IO Pads
- Standard Cells
- Macros + Halo
- Interconnects (utilization)



$$Utilization = \frac{Cell\ area + Macro\ area + Halo\ area}{Core\ area}$$

Rough Estimate based on:

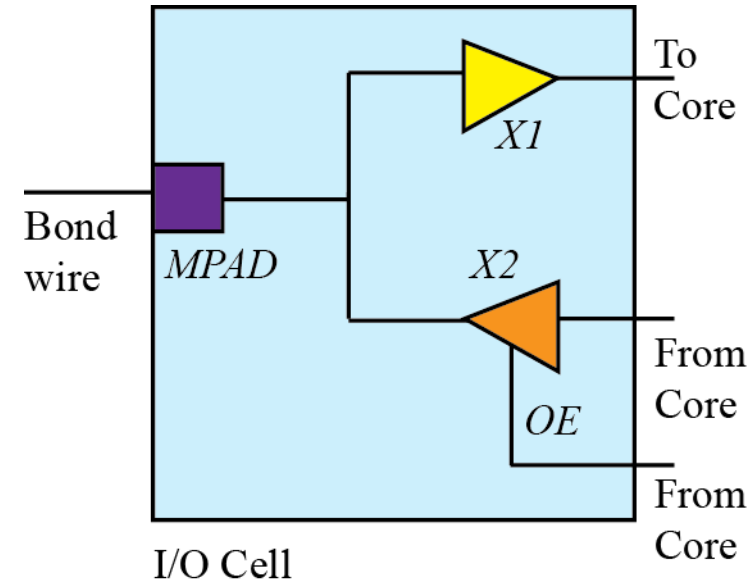
- Previous designs
- Available package size

IO Cells: Basics

- IO cells are special circuit elements through which a chip communicates with the external world
- Can be of type input, output, or bidirectional

Other functions of IO cells

- Drive capability
- Voltage transformation
- Protection against ESD (Electro-Static Discharge) [short high-voltage (several kilovolt) pulses]



IO Cells: Connection with Package

- Metallic pad (MPAD) is connected to the package pins using bonding wires
- Power pads: special cells that supply power to a chip
- IO Cells too need power and ground connections

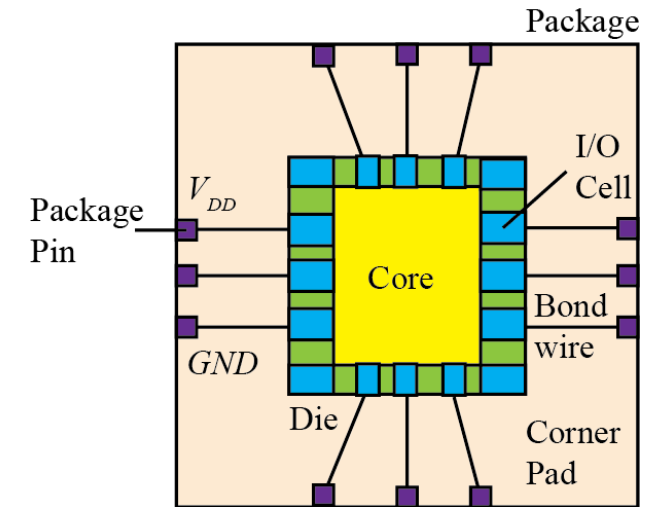
Number of power pads:

- Target level of internal voltages
- Current capacity of the power pads

Placement of IO Cells are guided by heuristics

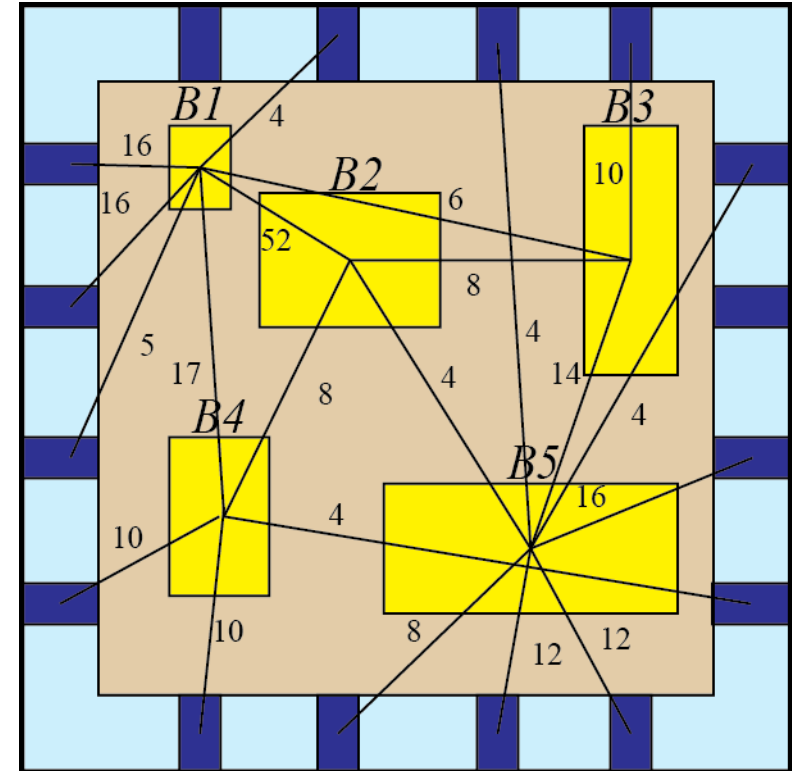
- Assign nearby positions to two primary inputs that jointly drive a multi-input logic gate
- Spread power-hungry I/O cells all over the die area to avoid creating voltage drop hotspots
- Avoid placing sensitive (such as clock signals) near IO cells

- Flip-chip technology offers more flexibility in placing macro cells.



Large Object (Macro) Placement

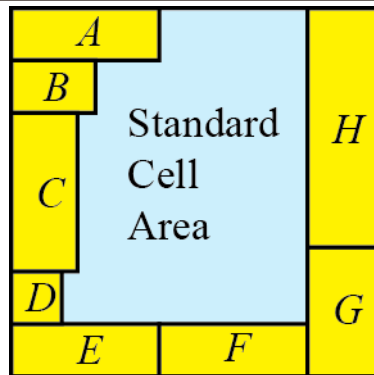
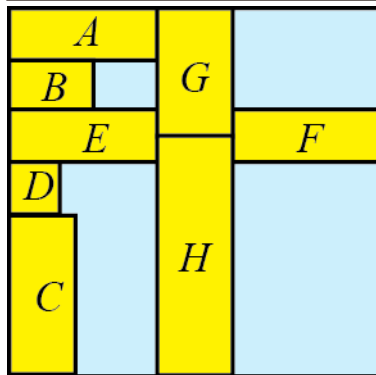
- Placement of large objects undertaken during floorplanning
 - Blocks (obtained in partition), RAM, analog blocks (such as PLL), etc.
 - Placement of standard will be undertaken during placement stage
-
- Initial macro placement is guided by connectivity
 - Strongly connected macros placed close together
 - Fly lines can guide the floorplan
 - Macros interacting with external world close to I/O pads
 - Predictive routing (rough routing) can be done to get an initial idea of problem in the floorplan



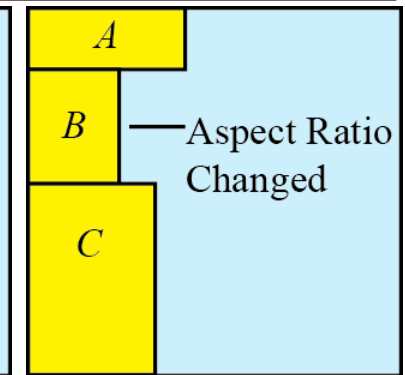
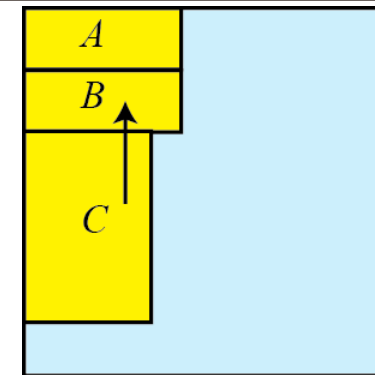
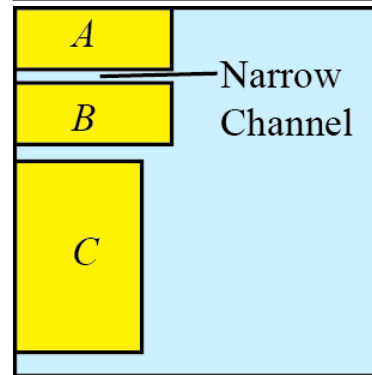
Macros Placement: Guidelines (1)

- Some guidelines can help in obtaining good floorplan

- Allot contiguous region for standard cells



- Avoid narrow channels between macros



References

- S. Saurabh, “Introduction to VLSI Design Flow”. Cambridge: *Cambridge University Press*, 2023.